Mathematics - Undergraduate Level Assessment 21 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A fair die is tossed 360 times. The probability that a six comes up on 70 or more of the tosses is
   (a) greater than 0.50
   (b) between 0.16 and 0.50
   (c) between 0.02 and 0.16
   (d) between 0.01 and 0.02

2. Let C be the circle in the yz-plane whose equation is (y - 3)^2 + z^2 = 1. If C is revolved around the z-axis, the surface generated is a torus. What is the equation of this torus?
   (a) $x^2 + y^2 + z^2 + 8 = 6$ y
   (b) $(x^2 + y^2 + z^2)^2 = 8 + 36(x^2 + z^2)$
   (c) $(x^2 + y^2 + z^2 + 8)^2 = 36(x^2 + z^2)$
   (d) $(x^2 + y^2 + z^2 + 8)^2 = 36(x^2 + y^2)$

3. Statement 1 | Suppose f : [a, b] is a function and suppose f has a local maximum. f'(x) must exist and equal 0? Statement 2 | There exist non-constant continuous maps from R to Q.
   (a) True, True
   (b) True, False
   (c) False, True
   (d) False, False

4. If A is the 2 by 2 matrix whose (i, j) entry is equal to i + j, and B is the 3 by 3 matrix whose (i, j) entry is equal to i + j, find the value of the sum detA + detB.
   (a) -2
   (b) -1
   (c) 0
   (d) 2

5. Statement 1 | R is a splitting field of some polynomial over Q. Statement 2 | There is a field with 60 elements.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: Only one positive integer k exists such that k! ends in exactly 99 zeros.
7. True or False: The probability that one integer will be the square of the other is 0.72.
8. True or False: The polynomial 4 x - 2 is irreducible over the field of rational numbers Q.
9. True or False: The cyclic subgroup of Z_24 generated by 18 has an order of 4.
10. True or False: The value of c in Z_3 that makes Z_3[x]/(x^3 + x^2 + c) a field is 1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. There is a committee composed of eight women and two men. When they meet, they sit in a row—the women in indistinguishable rocking chairs and the men on indistinguishable stools. How many distinct ways are there for me to arrange the eight chairs and two stools for a meeting?
12. Discuss the implications of Statement 1 that every group of order p^2 (where p is prime) is Abelian, and analyze Statement 2 regarding the conditions for a Sylow p-subgroup to be normal in a group G.

End of Paper